

Project Design Phase
Solution Architecture

Date	02 October 2026
Team ID	SWTID-2026-3464
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	4 Marks

Solution Architecture:

The **FitBuddy – AI Fitness Plan Generator using Gemini Models** uses a web-based architecture that connects the user interface, FastAPI backend, Gemini AI, and SQLite database. The architecture is designed to receive user fitness information, generate personalized workout plans, store relevant data, and provide updated recommendations based on user feedback.

Its main goals are to:

- **Provide a personalized fitness solution** based on the user's age, weight, fitness goal, and preferred workout intensity.
- **Connect the web interface with the FastAPI backend** to process user inputs, workout requests, and feedback.
- **Use Google Gemini AI** to generate personalized 7-day workout plans, nutrition guidance, recovery recommendations, and feedback-based plan updates.
- **Store and retrieve user information, workout plans, feedback, and workout history** using the SQLite database.
- **Provide a simple and interactive interface** through which users can view their workout plans and receive fitness guidance.

Example - Solution Architecture of FitBuddy – AI Fitness Plan Generator using Gemini Models

